Modelling Particle Production in Analog Expanding Universes with High-Performance Computing

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Introduction - Analog Universes

1. Inflation created particles

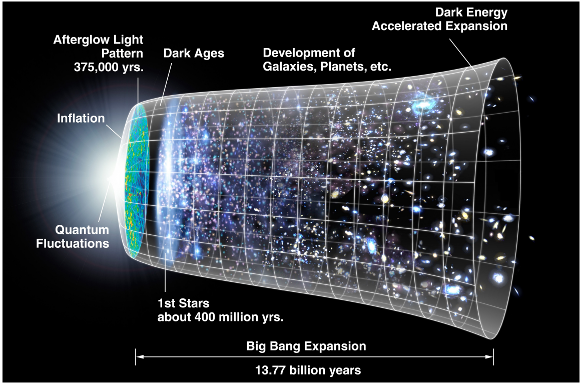

Figure: History of the universe highlighting period of rapid expansion (Source)

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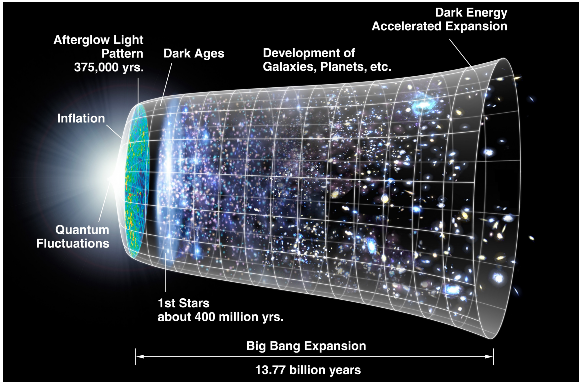
2. Cannot directly observe these particles



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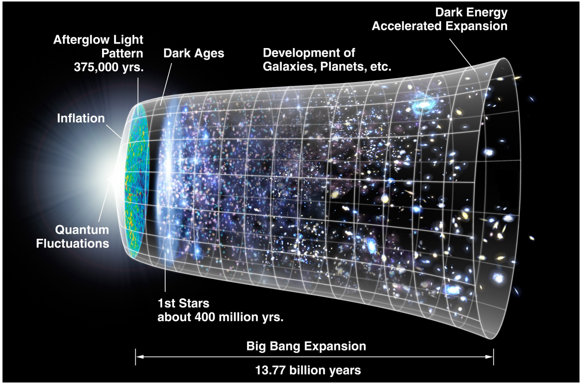

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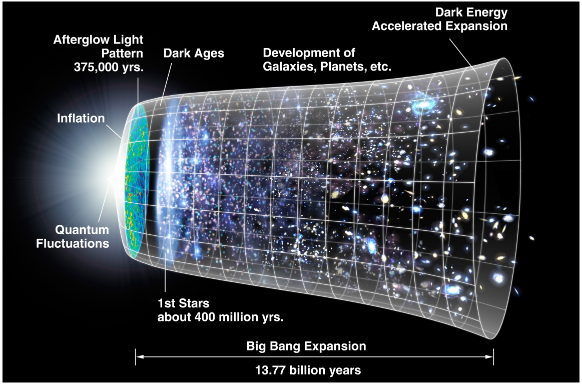
4. Can make gas in the lab $\implies$ can *effectively* observe inflation particles



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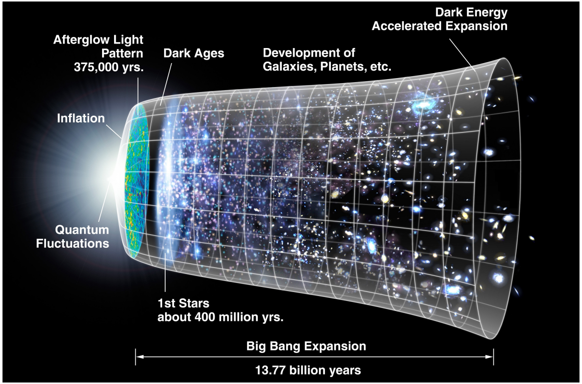
5. Lab based system = "analog universe"



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Introduction - The Story So Far

- Many investigations into analog universes

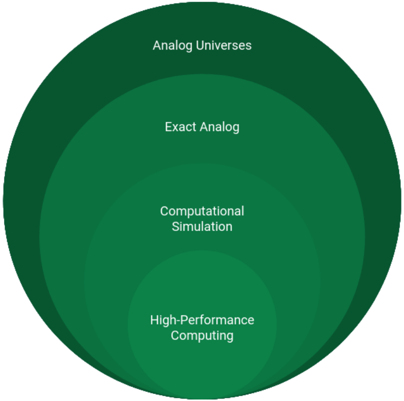

Figure: Subsets of increasingly specific literature

Introduction - The Story So Far

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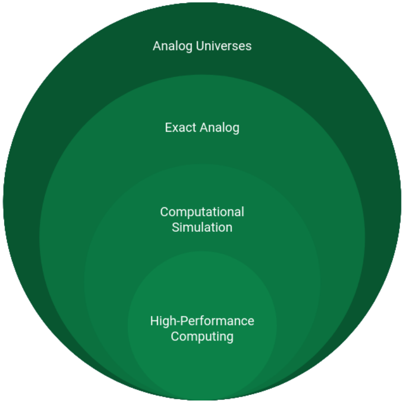

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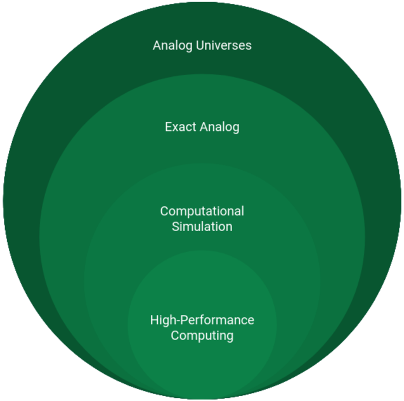

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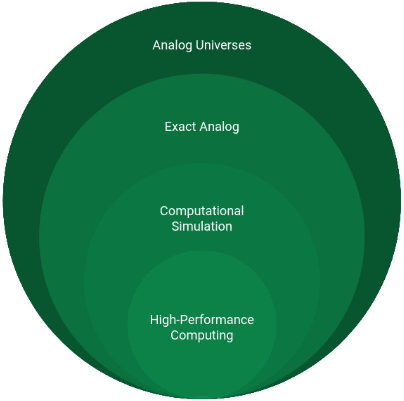
Introduction - The Story So Far

- Many investigations into analog universes
- Few into exact analogs
- Fewer into computational simulation
- Even fewer using high-performance computing



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Methods - Speed & Expansion

- Speed of sound in quantum gas =

  Speed of light in analog universe

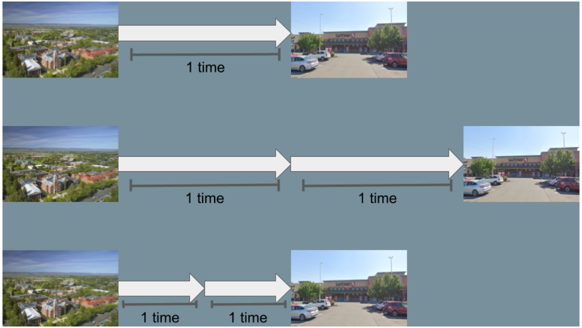

Figure: Travel time can be doubled by either doubling distance or halving speed

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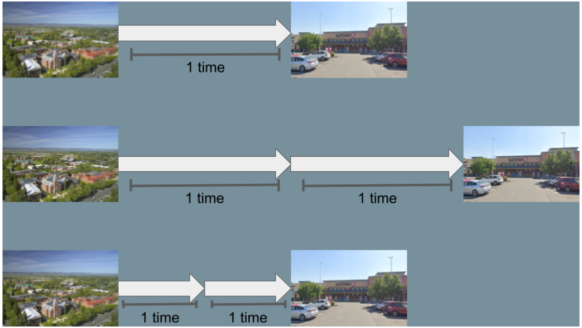

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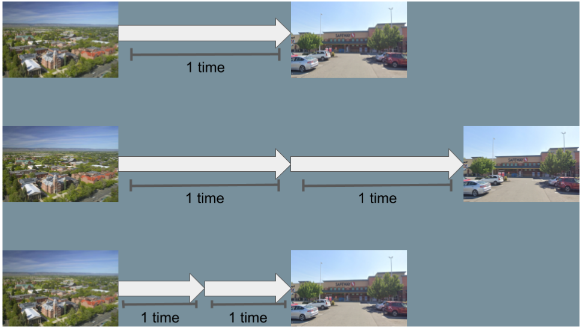
Methods - Speed & Expansion

- Speed of sound in quantum gas = Speed of light in analog universe
- Speed decrease & static volume $\iff$ constant speed & expanding volume
- Example: $t = d/v$
  - Distance to grocery store doubles $\implies$ travel time doubles
  - Speed to the grocery store halves $\implies$ travel time doubles



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Methods - The Field Equation

- One equation describes both ultra-cold quantum gases and inflation particles
- One equation for each $k < k_c$
- Universe expansion starts and stops at finite times

The Field Equation

$$\underbrace{\partial_\eta^2 \widetilde{\theta}_{\mathbf{k}} - c_0^2 k^2 \widetilde{\theta}_{\mathbf{k}}}_{\text{Wave}} - \underbrace{\frac{1}{2}\frac{\dot{a}(\eta)}{a(\eta)}\partial_\eta \widetilde{\theta}_{\mathbf{k}}}_{\text{Damping}} = 0$$

$$\widetilde{\theta}_{\mathbf{k}}(0) = \widetilde{\theta}_{\mathbf{k}0} \quad \partial_\eta \widetilde{\theta}_{\mathbf{k}}(0) = -U_0 n_{\mathbf{k}0}/\hbar$$

Methods - Solving The Field Equation

- Need to try to conserve energy

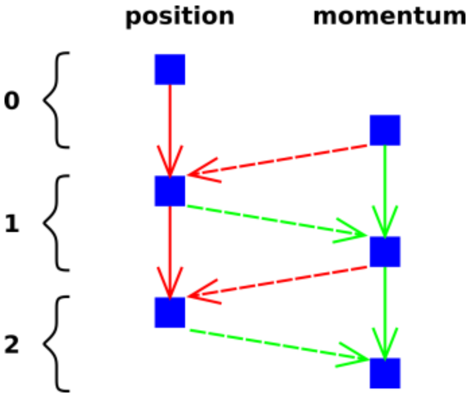

Figure: Symplectic integrator example: Velocity Verlet (Source)

Methods - Solving The Field Equation

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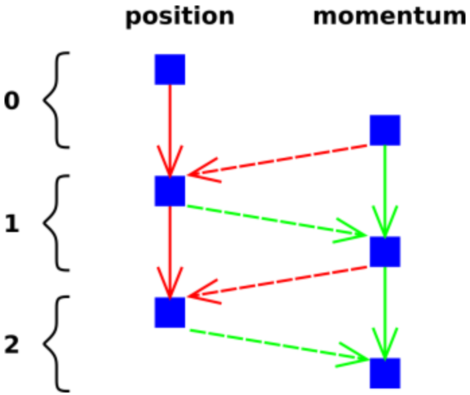

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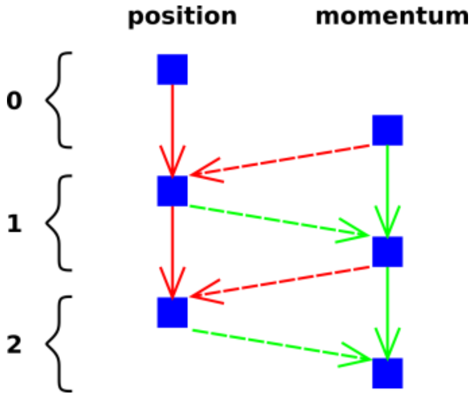

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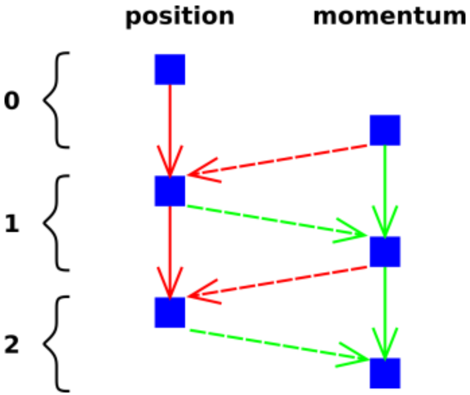
Methods - Solving The Field Equation

- Need to try to conserve energy
- Use *symplectic integrator*
- Use new momentum to calculate position
- More complex but more stable



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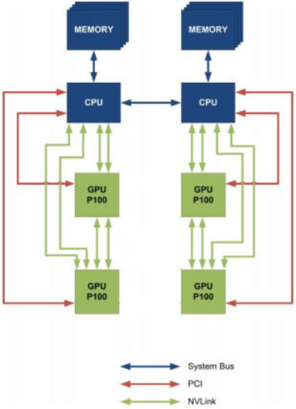

Figure: CPU to GPU and GPU to GPU interconnect using NVLink

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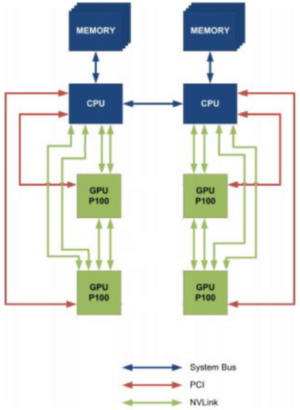

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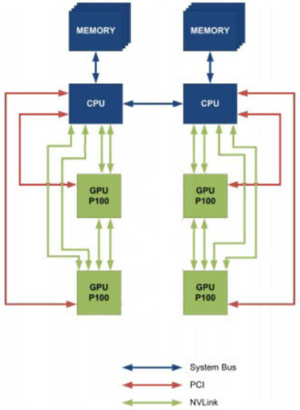

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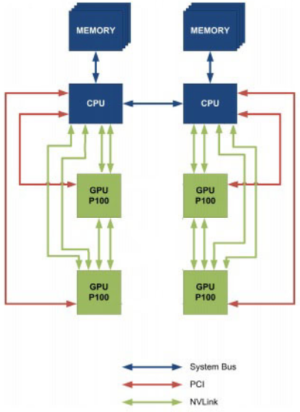
Methods - High-Performance Computing

- Equation for each $k < k_c$
- Many equations to solve
- Solve at the same time with multiple GPUs
- Use CWU's "Turing" supercomputer
- 4x4 NVIDA P100 GPUs

  - Each 83% performance of modern RTX 3060



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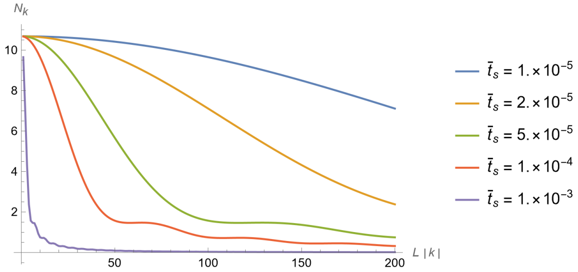
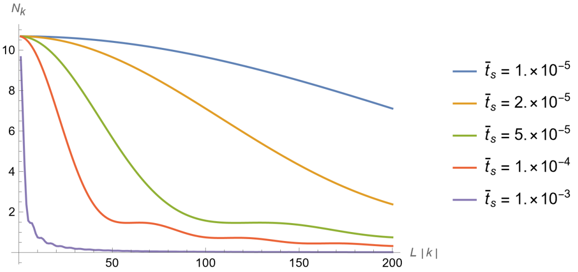


Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

Analytic Results - Simple Expansion

- Simple expansion: $a(t) = e^{-t/t_s}$
- Use Mathematica for analytic results



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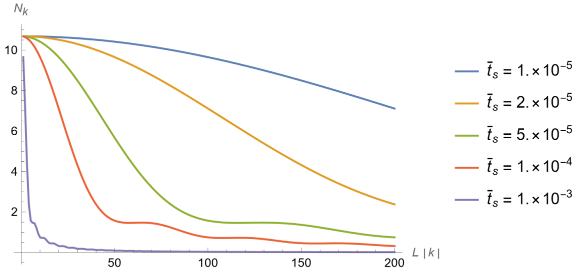

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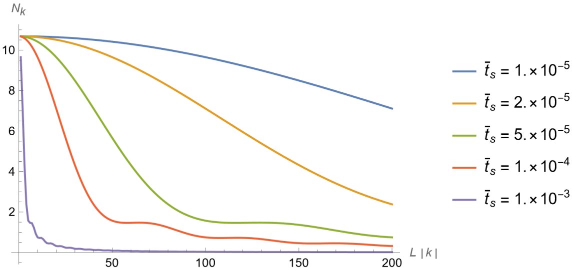

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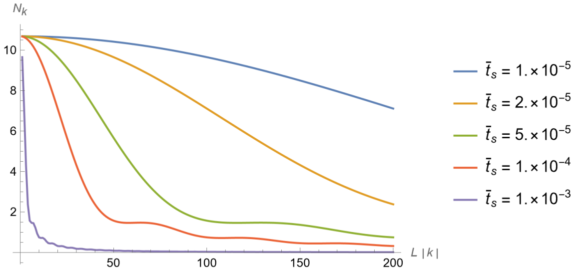
- Population drops off quickly with k



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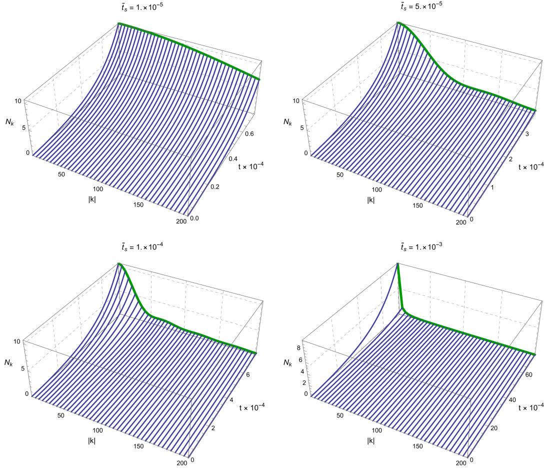

Figure: Momentum distribution throughout the expansion

Conclusion

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- Theory suggests inflation particles behave the same as ultra-cold quantum gases in the lab
- There are many equations that need to be solved with special methods to conserve energy
- Use high-performance techniques and hardware for reasonable execution time
- Expect the distribution of momenta to depend on the length of the expansion
- Expect the population to be high for low k and drop off quickly

Acknowledgements

- Dr. Andy Piacsek (CWU, Physics)
- Dr. Szilard VAJDA (CWU, Computer Science)
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- Dr. Brandon Peden (WWU, Physics & Astronomy)

Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): `github.com/nondairyneutrino/thesis`
- My website: `nathanwchapman.com`